Modelling Room Acoustics on a GPU

A new method that improves memory efficiency

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Dissertation Presentation

Introduction

This talk aims to answer the following questions:

- Modelling acoustics: what and why?
- What is the current state-of-the-art?
- What is the new method proposed here?
- What were the results when this method was used?
- How could these be improved?

Sound

What is it?

From Wikipedia

“In physics, sound is a vibration that propagates as a typically audible mechanical wave of pressure and displacement, through a transmission medium such as air or water.”

From my undergraduate degree

$$\nabla^2 P - \frac{1}{c^2}\frac{\partial^2 P}{\partial t^2} = 0 \tag{1}$$

Room acoustics

- The physical properties of rooms affects how we hear sound.
- Sound reverberates off walls/objects.
- Waves diffract around corners.
- Some rooms echo, some sound dry.
- Backwards: If we know physical properties can we derive the sound?

The wave equation

$$\frac{\partial^2 P}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 P}{\partial t^2} = 0$$

where P is pressure deviation and c is speed of sound in medium.

- Very familiar to those who have studied physics
- It, and its application to acoustics, are derivable from first principles.
- Solving for P gives us acoustic properties of a system (boundary conditions are very important!).
- Solving this analytically for complex systems is not possible.

FTDT

Finite-difference time-domain method

Finite difference methods are a way of approximating and discretising differential equations so that they can be solved numerically.

Start off by replacing derivative with finite difference:

$$\frac{df(u)}{du} \approx \frac{f(u+h) - f(u-h)}{2h} \tag{2}$$

$$\frac{d^2 f(u)}{du^2} \approx \frac{f(u+h) - 2f(u) + f(u-h)}{h^2} \tag{3}$$

Let h be the distance between samples (H) and P be a 3D grid

$$\delta_t^2 P_{l,m,p}^n \equiv \frac{1}{T^2}\left(P_{l,m,p}^{n+1} - 2P_{l,m,p}^n + P_{l,m,p}^{n-1}\right) \approx \frac{\partial^2 P}{\partial t^2} \tag{4}$$

$$\delta_x^2 P_{l,m,p}^n \equiv \frac{1}{H^2}\left(P_{l+1,m,p}^{n} - 2P_{l,m,p}^n + P_{l-1,m,p}^{n}\right) \approx \frac{\partial^2 P}{\partial x^2} \tag{5}$$

$$\delta_y^2 P_{l,m,p}^n \equiv \frac{1}{H^2}\left(P_{l,m+1,p}^{n} - 2P_{l,m,p}^n + P_{l,m-1,p}^{n}\right) \approx \frac{\partial^2 P}{\partial y^2} \tag{6}$$

$$\delta_z^2 P_{l,m,p}^n \equiv \frac{1}{H^2}\left(P_{l,m,p+1}^{n} - 2P_{l,m,p}^n + P_{l,m,p-1}^{n}\right) \approx \frac{\partial^2 P}{\partial z^2} \tag{7}$$

$$\frac{\partial^2 P}{\partial t^2} = c^2 \left(\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2} \right) \tag{8}$$

$$\begin{aligned} P^{n+1}_{l,m,p} - 2P^{n}_{l,m,p} + P^{n-1}_{l,m,p} = \frac{c^2 T^2}{H^2} \big(& P^{n}_{l+1,m,p} + P^{n}_{l-1,m,p} + P^{n}_{l,m+1,p} + P^{n}_{l,m-1,p} \\ & + P^{n}_{l,m,p+1} + P^{n}_{l,m,p-1} - 6P^{n}_{l,m,p} \big) \end{aligned} \tag{9}$$

Letting the Courant number be defined as

$$\lambda \equiv \frac{cT}{H} \tag{10}$$

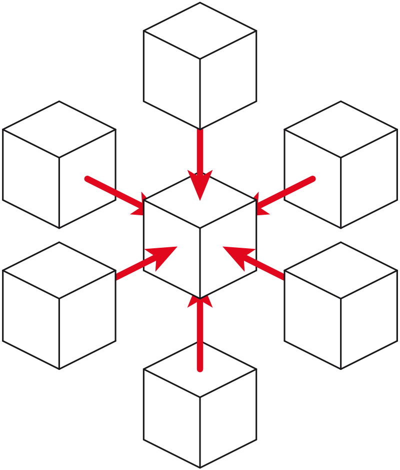
and our stencil

$$S \equiv P^n_{l+1,m,p} + P^n_{l-1,m,p} + P^n_{l,m+1,p} + P^n_{l,m-1,p} + P^n_{l,m,p+1} + P^n_{l,m,p-1} \tag{11}$$

gives a somewhat neater version of equation 9

$$P^{n+1}_{l,m,p} = (2 - 6\lambda^2)P^n_{l,m,p} + \lambda^2 S - P^{n-1}_{l,m,p} \tag{12}$$

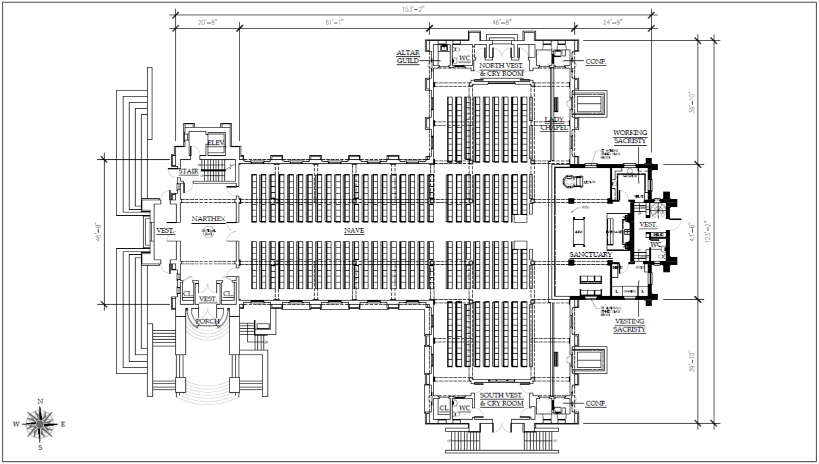
Stencil



Memory considerations

For stencilling operations, memory access is the bottleneck.

- Accurate models need a lot of points to model a room.
- Wavelength of sound at 20kHz is 17mm.
- High memory bandwidth means GPUs are used - limited memory.
- Standard 3d arrays are cuboid.
- Due to memory layout, neighbouring points in space will always be far apart in memory.



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HOLY CROSS CHURCH

RUMSON, NEW JERSEY

Main Floor Plan
1/16" = 1'-0"
December 29, 2011
Sheet 4 of 21

The Proposal

Use an array of structs instead!

- Store enough points for CUDA kernel to operate on.
- Only store structs with at least one point inside the room.
- Experiment with different sized arrays within the struct to see if caching benefits can be yielded.

The Struct

Listing 1: The block struct

```
typedef real bl_array[Bz][By][Bx];
struct block {
    int up;
    int down;
    int left;
    int right;
    int fore;
    int aft;
    bl_array u;
    bl_array u1;
    char k[Bz][By][Bx];
};
^^I
```

Converting between representations

Going from bounding box to array of structs

- Implement struct in 2d, then 3d
- Implement algorithm to convert from bounded array to array of structs.
- See if this uses less memory (it does!)
- See if this method is intolerably slow in CUDA.
- See if results are **correct**.
- Experiment with different sub-array sizes to optimise for
  - memory usage
  - caching efficiency

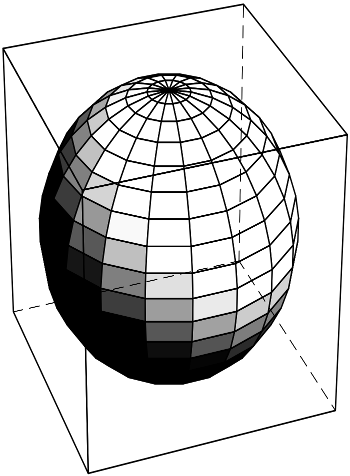

Figure: Computer graphic showing a sphere of unit diameter within a cube of unit side.

The End

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